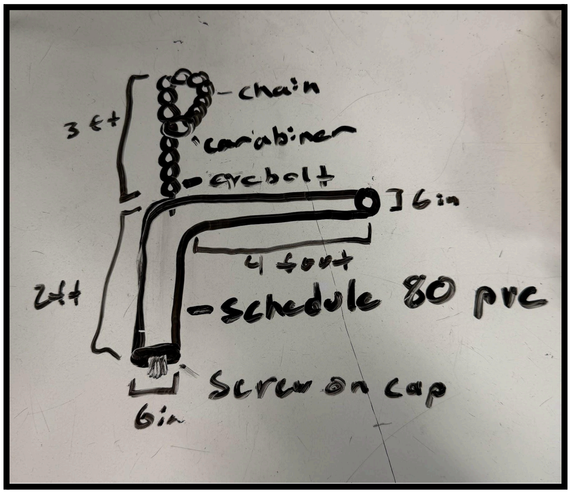
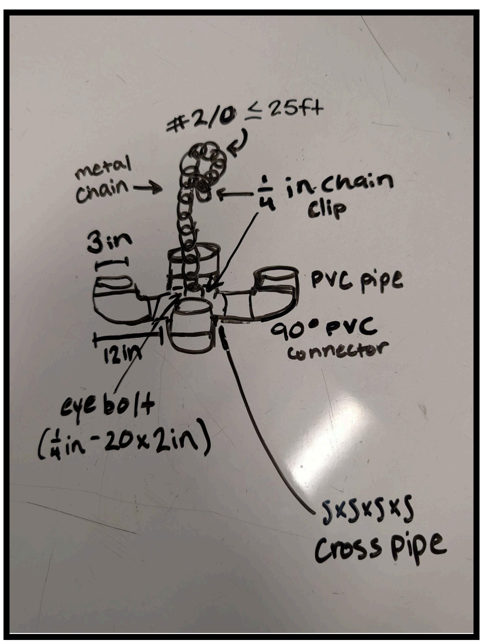
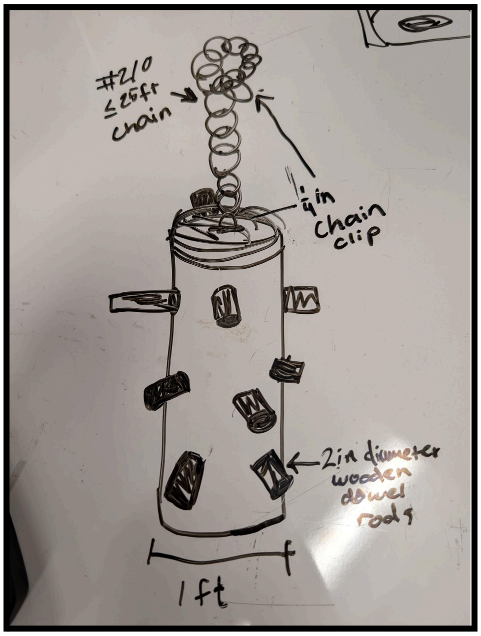
Element D: Design Concept Generation, Analysis, and Selection



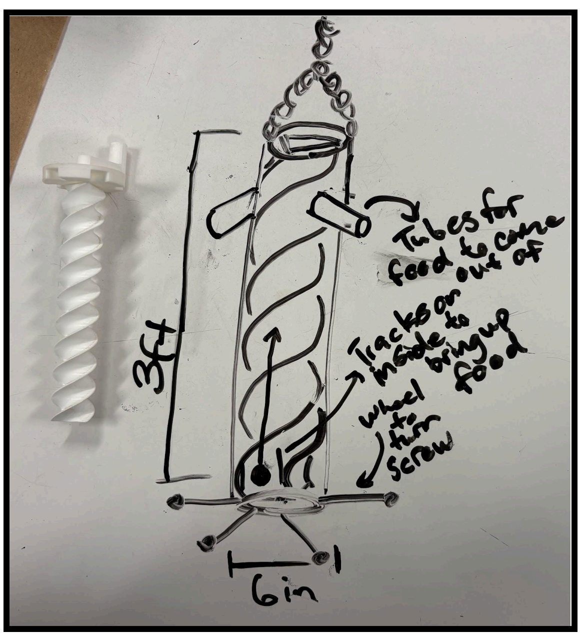
DESIGN 1: Boomerang Feeder: A hanging L shaped pvc pipe that includes one capped end and a longer opened end. The toy would have food placed into the capped in and the pipe would need to be dipped so that the food could fall out the opened end. This design is easy to maneuver for the zookeepers, safe for the leopard, easy to clean and prep, and only uses materials that are already cleared for use in the exhibit.



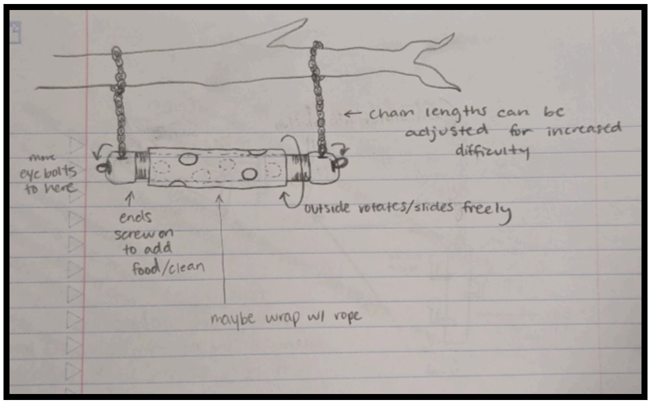
DESIGN 2: The Chandelier: Another chandelier design that would feature four conjoined pvc pipes all with openings out the top. The food would be placed in the center and would have to be maneuvered out of the openings most likely by means of smacking the chandelier. Similar to design 1, this design would be easy to carry, easy to set up, challenging for the leopard to maneuver, but it may prove a little difficult to clean. It is also designed with pre-approved materials for the leopards exhibit.



DESIGN 3: Stick Stack Feeder. This design includes a long pipe (whether PVC or trash can tube) with long poles that stick out of either end of the pipe. They are able to rotate and be pushed or pulled out freely from the holes in the main pipe. This design allows food to drop out of the bottom of the contraption when the poles are shoved or pulled out of their holes. The main setback of this design was the uncertainty when it comes to the food falling from the top since there aren't set borders that prohibit the prize from falling through the bars.



Design 4: Spiral Feeder: Uses an Archimides screw to deliver food to the top of a 3 foot long pvc pipe. Tubes at the top would be placed that food can fall out of once it reaches the top. The leopard would be able to manipulate the screw from the bottom of the feeder, where 6 inch wooden pegs would be placed. The leopard can rotate these to push the food up and eventually out of the tube. This design does have a very good puzzle mechanic but it could prove too difficult for the leopard to solve. Also, the archimides screw itself would be extremely difficult to produce, maintain, and clean.



Design 5: The Hole In One: This design was chosen for its combination of variability, mental stimulation, and ease of use. With this design, there is a rotating PVC pipe of larger diameter than the PVC on the interior. Both PVC pipes have multiple holes in which food can be hidden, and the leopard must align the outside PVC holes with the holes set on the inside in order to reach the food. The outer PVC may be wrapped in rope in order to cater to the leopard's grip and natural aesthetic of the exhibit. The two end caps can be screwed on and off, making the cleaning process simple and straightforward. The contraption can also hang from differing heights by adjusting the length of the two chains.

Decision Matrix:

We gave each high-ranking criteria a value of 1, and we had no low value criteria. (The criteria are ranked on a scale of 1-5 where 5 is the best score)

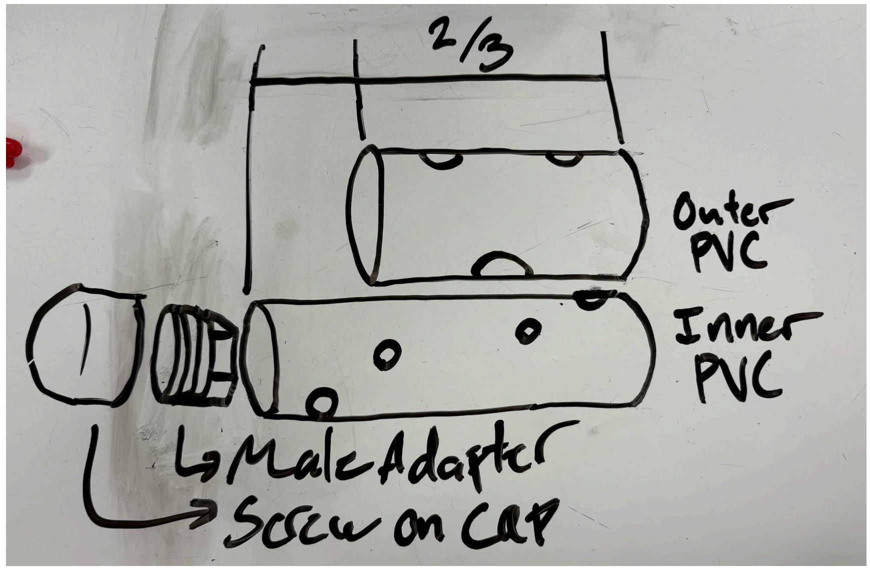
Design #	Safety	Cleanability	Difficulty in Solving	Durability	Longevity	Mobility	Total Score
1 Boomerang Feeder	3	2	3	4	4	5	21
2 Chandelier	4	1	1	3	2	2	13
3 Stick-Stack Feeder	1	5	3	1	3	1	14
4 Spiral Feeder	2	2	5	2	1	3	15
5 Whole lotta fun	3	4	4	5	5	4	25

The Whole lotta fun feeder had the most points based off of the decision matrix.

Design Blueprint:

Manufacturing Method:

1. Obtain all parts and prep the chain by splitting the 25’ into two sections.
2. Take the PVC pipe and drill 4, 3” holes into the sides. Make these holes on various sides of the pipe and at equal variables along the pipe. Please note that these holes will have to line up with the outer holes as well.



3. Take the aluminum pipe and drill aligning holes into it. Make these holes 3.5” to allow for a little more movement within the holes.
4. Sand both ends of the PVC pipe and clean any residue to prep for the PVC glue.
5. Sand both PVC adapters and clean any residue to prep for the PVC glue.
6. Apply PVC glue to both ends of the PVC pipe and both PVC adapters.
7. Attach one adapter onto the PVC pipe and slide the aluminum pipe over the open end. Now attach the second adapter. Let both of these dry.
8. Screw both sets of eye bolts into the caps.
9. Place the two carabiners onto the two chains. Attach the other end of the chains too the eye bolt.

Parts list/Cost:

1. 3’, 4”, Schedule 40, PVC pipe. $0, obtained from the workshop closet for free.
2. 3’, 5”, T6, Aluminium Tube, $96.78
3. 2X 4” PVC Male Adapter, $24.98
4. 2X 4” PVC Screw on Cap, $27.26
5. 2X ¼” Screw Carabiner, $8.7
6. 4X 3/16” Eye bolt, $3.14
7. 25’ Link Chain, $25.00
8. PVC Glue, $0, obtained from the workshop closet for free.

Gantt Chart (Big Ideas): Leopard Enrichment Project Ghant Chart

https://docs.google.com/spreadsheets/d/1QbrznbjKvuCaOUWByWVETKUiV-64_b2evjGqcIESbU/edit?usp=sharing